

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 11

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Activity 1

Write a menu driven shell script to perform the following tasks.

[1] Show Today's date/time

[2] Show files in current directory

[3] Show calendar

[4] Start editor to write letters

[5] Exit/Stop

It will execute repeatedly until user selects exit (option 5) from the menu.



```
GNU nano 7.2 a1.sh
#!/bin/bash

while true; do
  echo "=====
  echo "  MAIN MENU  "
  echo "=====
  echo "[1] Show Today's date/time"
  echo "[2] Show files in current directory"
  echo "[3] Show calendar"
  echo "[4] Start editor to write letters"
  echo "[5] Exit/Stop"
  echo "=====
  read choice

  case $choice in
    1) echo "Today's Date/Time: $(date)" ;;
    2) echo "Files in current directory:"; ls ;;
    3) cal ;;
    4) nano letter.txt ;;
    5) echo "Goodbye!"; exit 0 ;;
    *) echo "Invalid option. Please try again." ;;
  esac

  echo ""
done
```

```
kamshiga@kamshiga-VirtualBox: ~  
kamshiga@kamshiga-VirtualBox:~$ sudo apt install ncal  
[sudo] password for kamshiga:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following NEW packages will be installed:  
  ncal  
0 upgraded, 1 newly installed, 0 to remove and 321 not upgraded.  
Need to get 21.0 kB of archives.  
After this operation, 59.4 kB of additional disk space will be used.  
Get:1 http://lk.archive.ubuntu.com/ubuntu noble/universe amd64 ncal amd64 12.1.8 [21.0 kB]  
Fetched 21.0 kB in 2s (12.9 kB/s)  
Selecting previously unselected package ncal.  
(Reading database ... 190953 files and directories currently installed.)  
Preparing to unpack .../archives/ncal_12.1.8_amd64.deb ...  
Unpacking ncal (12.1.8) ...  
Setting up ncal (12.1.8) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
kamshiga@kamshiga-VirtualBox:~$  
kamshiga@kamshiga-VirtualBox:~$
```

When option 3 was selected, the error `cal: command not found` appeared. This is because the `cal` command is not pre-installed in this Ubuntu system. To fix this, the following command was run to install it:

```
sudo apt install ncal
```

After installation, option 3 worked correctly and displayed the calendar.

```
kamshiga@kamshiga-VirtualBox: ~  
kamshiga@kamshiga-VirtualBox:~$ nano a1.sh  
kamshiga@kamshiga-VirtualBox:~$ chmod +x a1.sh  
kamshiga@kamshiga-VirtualBox:~$ ./a1.sh  
=====
```

```
MAIN MENU  
=====
```

```
[1] Show Today's date/time  
[2] Show files in current directory  
[3] Show calendar  
[4] Start editor to write letters  
[5] Exit/Stop  
=====
```

```
1  
Today's Date/Time: Mon Apr 27 08:45:58 PM +0530 2026  
=====
```

```
MAIN MENU  
=====
```

```
[1] Show Today's date/time  
[2] Show files in current directory  
[3] Show calendar  
[4] Start editor to write letters  
[5] Exit/Stop  
=====
```

```
2  
Files in current directory:
```

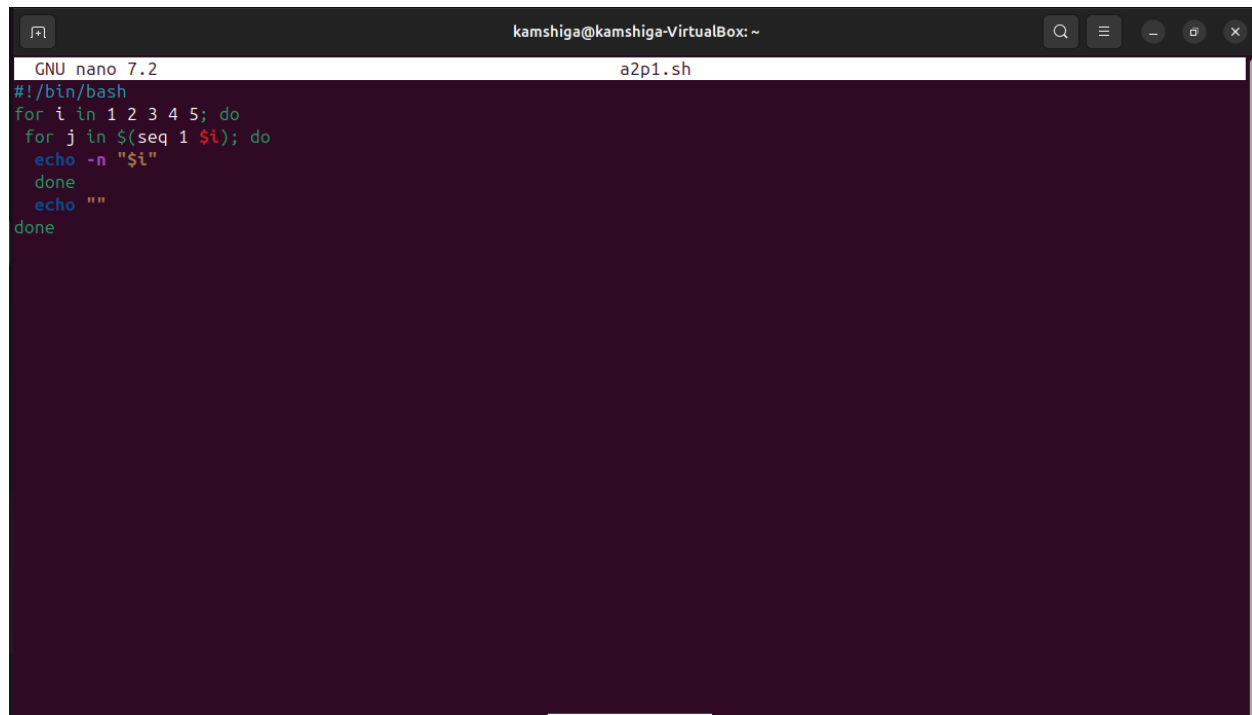
a1	a3	ac4.c	act4	activity1.c	c2	Documents	k3.c	que1	t2	ta3
a1.c	a3.c	ac5	act4.c	b1	c2.c	Downloads	k4	que1.c	t22	ta3.c
a1.sh	a3.sh	ac5.c	act5	b1.c	c3	e1.c	k4.c	que2	t22.c	ta4
a2	a4.sh	ac6	act5.c	b2	c3.c	first	k4.c.save	que2.c	t22.c.save	ta4.c

```
kamshiga@kamshiga-VirtualBox: ~
MAIN MENU
=====
[1] Show Today's date/time
[2] Show files in current directory
[3] Show calendar
[4] Start editor to write letters
[5] Exit/Stop
=====
Invalid option. Please try again.
=====
MAIN MENU
=====
[1] Show Today's date/time
[2] Show files in current directory
[3] Show calendar
[4] Start editor to write letters
[5] Exit/Stop
=====
3
    April 2026
Su Mo Tu We Th Fr Sa
      1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30
=====
```

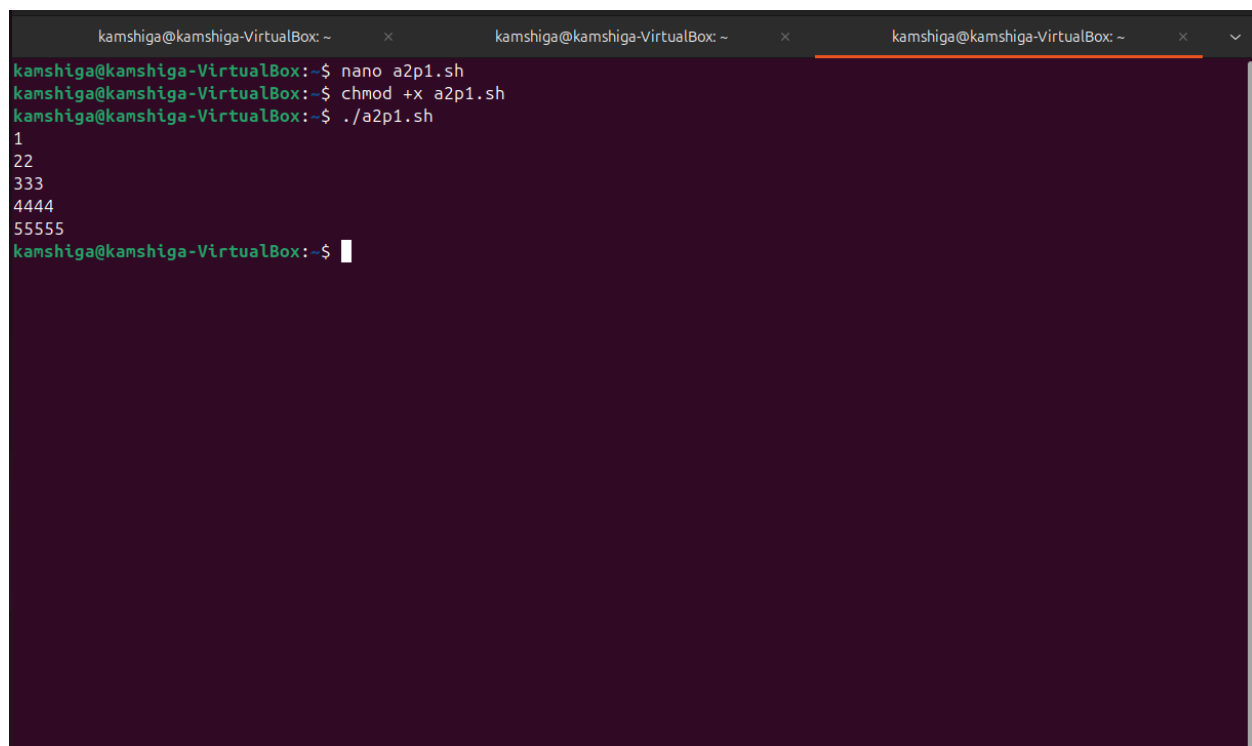
```
=====
MAIN MENU
=====
[1] Show Today's date/time
[2] Show files in current directory
[3] Show calendar
[4] Start editor to write letters
[5] Exit/Stop
=====
4
=====
MAIN MENU
=====
[1] Show Today's date/time
[2] Show files in current directory
[3] Show calendar
[4] Start editor to write letters
[5] Exit/Stop
=====
5
Goodbye!
kamshiga@kamshiga-VirtualBox:~$
```

Activity 2

Pattern 1

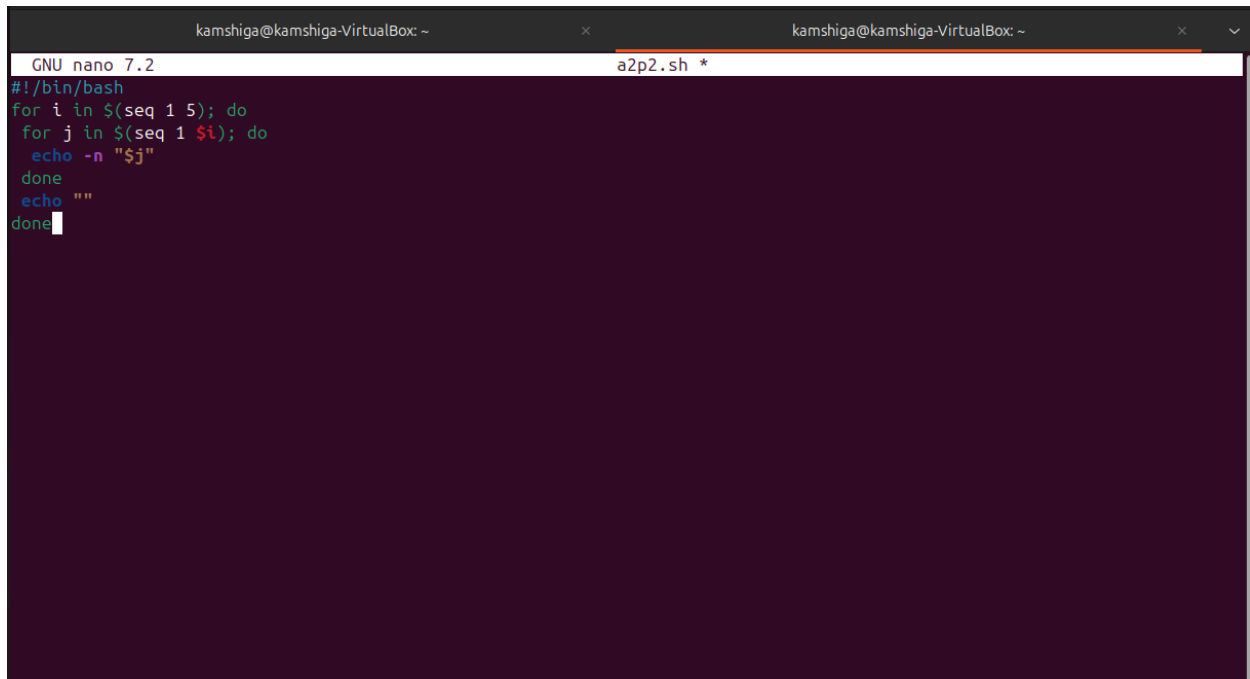


```
GNU nano 7.2 a2p1.sh
#!/bin/bash
for i in 1 2 3 4 5; do
  for j in $(seq 1 $i); do
    echo -n "$i"
  done
  echo ""
done
```



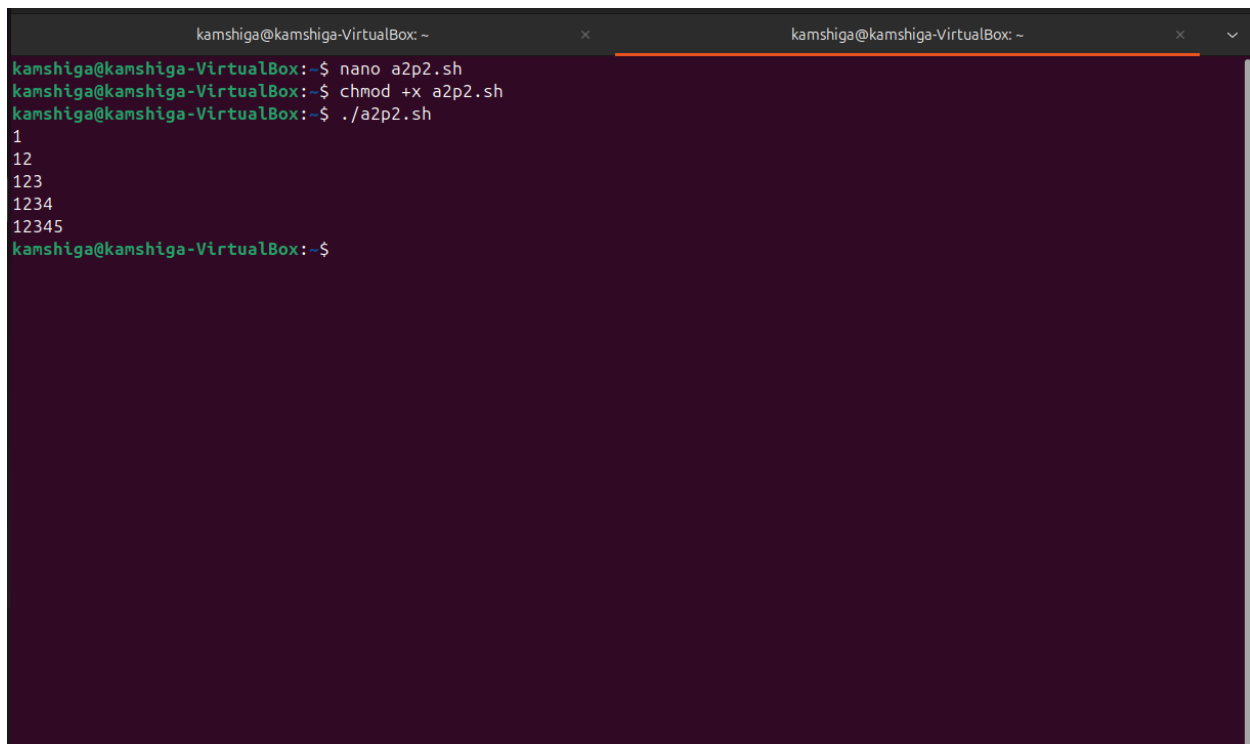
```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox: ~$ nano a2p1.sh
kamshiga@kamshiga-VirtualBox: ~$ chmod +x a2p1.sh
kamshiga@kamshiga-VirtualBox: ~$ ./a2p1.sh
1
22
333
4444
55555
kamshiga@kamshiga-VirtualBox: ~$
```

Pattern 2



The screenshot shows a terminal window with two tabs. The active tab is titled 'a2p2.sh *'. Inside the terminal, the GNU nano 7.2 editor is open, displaying the following script content:

```
#!/bin/bash
for i in $(seq 1 5); do
  for j in $(seq 1 $i); do
    echo -n "$j"
  done
  echo ""
done
```



The screenshot shows the same terminal window after the script has been executed. The terminal history shows the following commands and output:

```
kamshiga@kamshiga-VirtualBox: ~$ nano a2p2.sh
kamshiga@kamshiga-VirtualBox: ~$ chmod +x a2p2.sh
kamshiga@kamshiga-VirtualBox: ~$ ./a2p2.sh
1
12
123
1234
12345
kamshiga@kamshiga-VirtualBox: ~$
```

Pattern 3

```
kamshiga@kamshiga-VirtualBox: ~  
GNU nano 7.2 a2p3.sh  
#!/bin/bash  
n=5  
for i in $(seq 1 $n); do  
  for j in $(seq 1 $i); do  
    if [ $i -eq $n ]; then  
      echo -n "| "  
    elif [ $j -eq $i ]; then  
      echo -n "|_"  
    else  
      echo -n "| "  
    fi  
  done  
  echo ""  
done
```

```
kamshiga@kamshiga-VirtualBox: ~  
kamshiga@kamshiga-VirtualBox:~$ nano a2p3.sh  
kamshiga@kamshiga-VirtualBox:~$ chmod +x a2p3.sh  
kamshiga@kamshiga-VirtualBox:~$ ./a2p3.sh  
|_  
| |_  
| | |_  
| | | |_  
| | | | |  
kamshiga@kamshiga-VirtualBox:~$
```

Pattern 4

```
kamshiga@kamshiga-VirtualBox: ~  
GNU nano 7.2 a2p4.sh  
#!/bin/bash  
n=5  
#Upper half(1 to 5)  
for i in $(seq 1 $n); do  
  for j in $(seq 1 $i); do  
    echo -n "*" "  
  done  
  echo "  
done  
#Lower half (5 down 2)  
for i in $(seq $n -1 2); do  
  for j in $(seq 1 $i); do  
    echo -n "*" "  
  done  
  echo "  
done
```

```
kamshiga@kamshiga-VirtualBox: ~  
kamshiga@kamshiga-VirtualBox:~$ nano a2p4.sh  
kamshiga@kamshiga-VirtualBox:~$ chmod +x a2p4.sh  
kamshiga@kamshiga-VirtualBox:~$ ./a2p4.sh  
*  
* *  
* * *  
* * * *  
* * * * *  
* * * * *  
* * * *  
* * *  
* *  
* *  
kamshiga@kamshiga-VirtualBox:~$
```


Activity 3

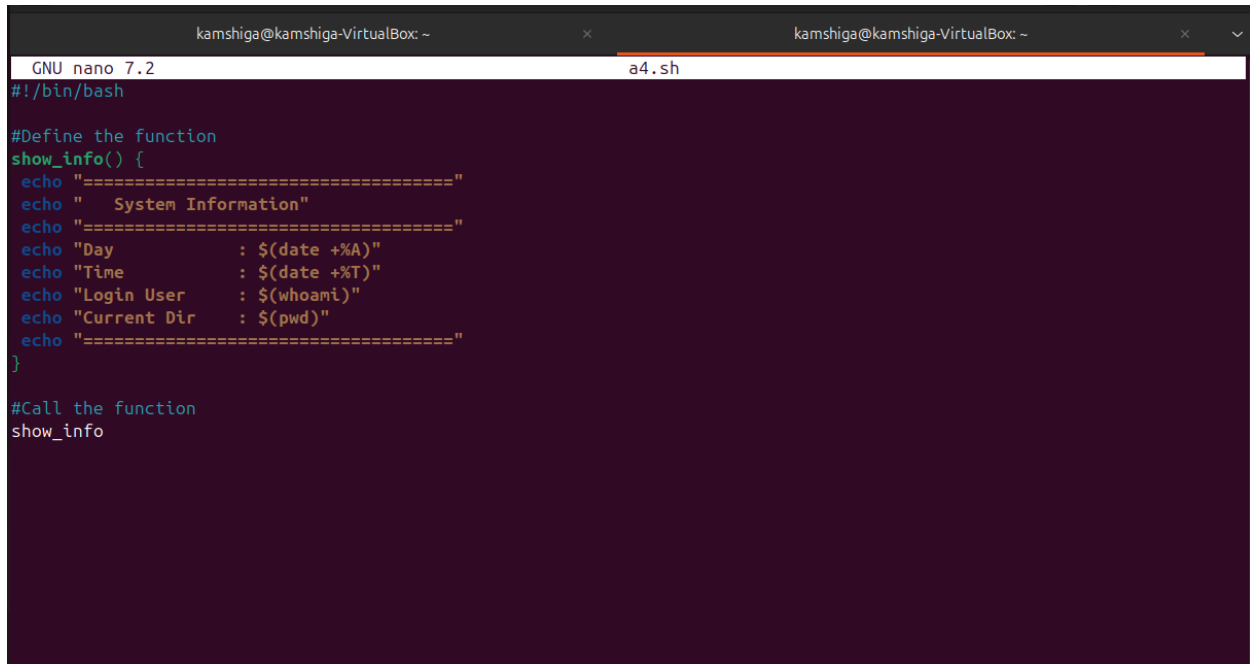
Write Script to find out biggest number from given three numbers. Numbers are supplies as command line argument. Print error if sufficient arguments are not supplied.

```
kamshiga@kamshiga-VirtualBox: ~  
GNU nano 7.2 a3.sh  
#!/bin/bash  
  
#Check if exactly 3 arguments are provided  
if [ $# -ne 3 ]; then  
    echo "Error: Please supply exactly 3 numbers as arguments."  
    echo "Usage: $0 num1 num2 num3"  
    exit 1  
fi  
  
a=$1  
b=$2  
c=$3  
  
#Find the biggest  
if [ $a -ge $b ] && [ $a -ge $c ]; then  
    biggest=$a  
elif [ $b -ge $a ] && [ $b -ge $c ]; then  
    biggest=$b  
else  
    biggest=$c  
fi  
  
echo "Numbers: $a, $b, $c"  
echo "Biggest number is: $biggest"
```

```
kamshiga@kamshiga-VirtualBox: ~  
kamshiga@kamshiga-VirtualBox:~$ nano a3.sh  
kamshiga@kamshiga-VirtualBox:~$ chmod +x a3.sh  
kamshiga@kamshiga-VirtualBox:~$ ./a3.sh 10 45 23  
Numbers: 10, 45, 23  
Biggest number is: 45  
kamshiga@kamshiga-VirtualBox:~$
```

Activity 4

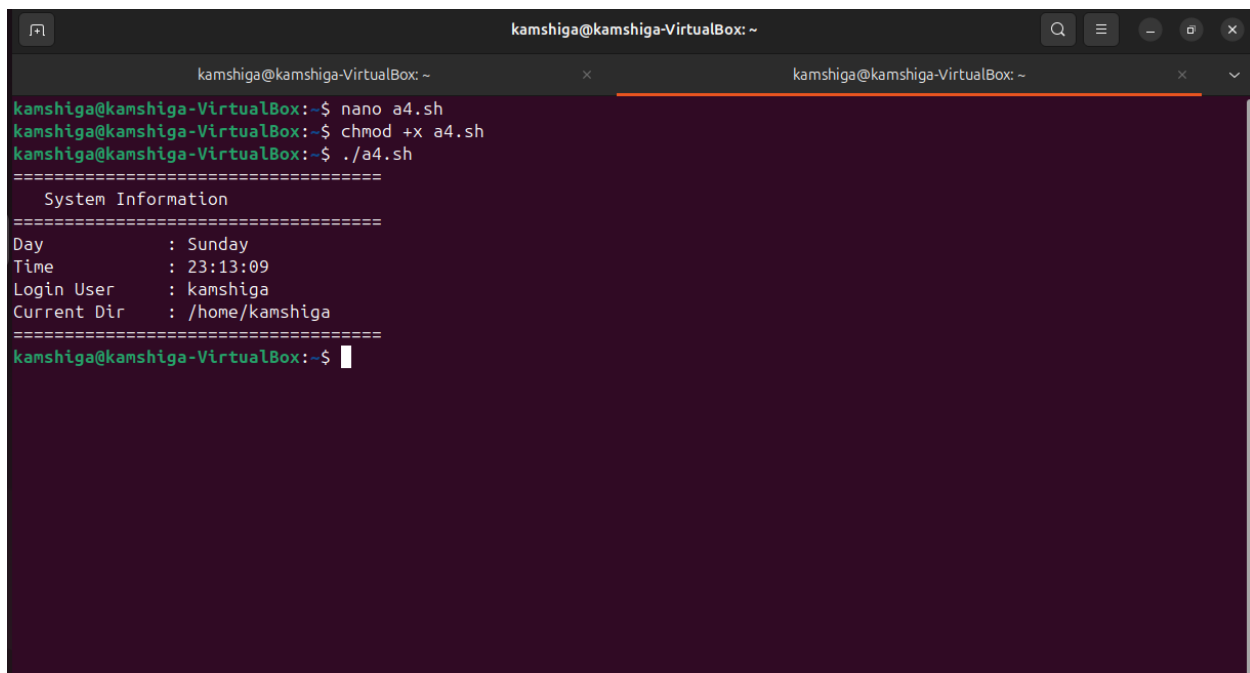
Write a function for bash shell to display the day, time, login and current directory.



```
GNU nano 7.2 a4.sh
#!/bin/bash

#Define the function
show_info() {
  echo "======"
  echo "  System Information"
  echo "======"
  echo "Day       : $(date +%A)"
  echo "Time      : $(date +%T)"
  echo "Login User : $(whoami)"
  echo "Current Dir : $(pwd)"
  echo "======"
}

#Call the function
show_info
```



```
kamshiga@kamshiga-VirtualBox: ~
kamshiga@kamshiga-VirtualBox:~$ nano a4.sh
kamshiga@kamshiga-VirtualBox:~$ chmod +x a4.sh
kamshiga@kamshiga-VirtualBox:~$ ./a4.sh
======"
  System Information
======"
Day       : Sunday
Time      : 23:13:09
Login User : kamshiga
Current Dir : /home/kamshiga
======"
kamshiga@kamshiga-VirtualBox:~$
```